Minh-Tâm Quang Trinh

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Citizenship

American

Current Research

algebraic geometry, combinatorics, quantum algebra, representation theory

Employment

Assistant Professor (tenure-track). Howard University 2025–

Gibbs Assistant Professor. Yale University 2024–25

CLE Moore Instructor. Massachusetts Institute of Technology 2020–24
NSF Mathematical Sciences Postdoctoral Research Fellow (2020–2023)
— *Sponsoring Scientist:* Zhiwei Yun

Education

PhD in Mathematics. University of Chicago 2014–20
— *Primary Advisor:* Châu Ngô (= Ngô Bảo Châu)
— *Secondary Advisor:* Victor Ginzburg
William Rainey Harper Dissertation Fellow

AB in Mathematics magna cum laude. Princeton University 2010–14

Publications and Preprints

— Free Immersions in Critical Dimension from Lie Groups
Preprint. arXiv:2609.24777

— Haiman's Conjecture and Springer's Representations
Preprint. arXiv:2605.20131

— Partial Resolutions and Noncrossing Combinatorics
Preprint. arXiv:2601.17293
\+ N. Williams

— Level-Rank Dualities from Φ-Cuspidal Pairs and Affine Springer Fibers
Preprint. arXiv:2311.17106
\+ T. Xue

— Unipotent Elements and Kálmán–Serre Duality
Preprint. arXiv:2210.09051

— From the Hecke Category to the Unipotent Locus
Preprint. arXiv:2106.07444

— The Hilb-vs-Quot Conjecture
J. angew. reine Math. (Crelle), **828** (2025), 83–126.
+ O. Kivinen

— Simple Braids Tend Toward Positive Entropy
J. Knot Theory Ramifications, **33**(11) (2024), 2450034, 12 pp.
+ L. Robitaille

— Rational Noncrossing Coxeter–Catalan Combinatorics
Proc. Lond. Math. Soc. **129**(4) (2024), e12643, 50 pp.
+ P. Galashin, T. Lam, N. Williams

— Algebraic Braids and Geometric Representation Theory
PhD thesis. University of Chicago. ProQuest LLC (2020), 239 pp.

— Extremal Primes for Elliptic Curves
J. Num. Theory, **164** (2016), 282–298.
+ K. James, B. Tran, P. Wertheimer, D. Zantout

— Zeros of Dirichlet L-Functions over Function Fields
Comm. Num. Theory Phys., **8**(3) (2014), 511–539.
+ J. Andrade, S. J. Miller, K. Pratt

Extended Abstracts

— Level-Rank Dualities for Finite Reductive Groups
To appear in *Math. Res. Lett.* (2026). arXiv:2508.07051
+ T. Xue

— Cell Decompositions of Hecke Traces and Link Polynomials
Proceedings of the 37th International Conference on Formal Power Series and Algebraic Combinatorics (FPSAC), Sapporo, 2025. *Sém. Lothar. Combin.*, **93B** (2025), Paper No. 90.
+ N. Williams

Non-Research Publications

— The Hodge Theory of Soergel Bimodules
Chapter 18 in *Introduction to Soergel Bimodules*. Ed. B. Elias, S. Makisumi, U. Thiel, G. Williamson. RSME Springer Series, **5** (2020), 347–367.
+ L. Taylor

Invited Talks

— KIAS–PKU–SKKU Combinatorics and Representation Theory Seminar. 12/11/26.
— UT-Austin Geometry Seminar. 12/3/26.
— SIAM TXLA Sectional Meeting. Minisymposium on Algebraic and Geometric Structures in Discrete Mathematics. 20-minute talk. SMU. 10/31/26.

— Workshop II: Contact Geometry, Cluster Algebras, and Skein Theory. Quantum Topology, Character Varieties and Low-Dimensional Geometry. Institute for Pure & Applied Mathematics (IPAM). 10/27/26.
— AMS Fall Eastern Sectional Meeting. Special Session on Categorification, Low-Dimensional Topology, and Representation Theory. 50-minute talk. GWU. 10/4/26.
— AMS Fall Eastern Sectional Meeting. Special Session on Knots and Braids. 20-minute talk. GWU. 10/3/26.
— MIT Lie Groups Seminar. 4/29/26.
— Knots in Washington 53. 4/25/26. Plenary talk.
— UT-Dallas Algebra and Combinatorics Seminar. 3/19/26.
— GWU–Howard Combinatorics & Algebra Seminar. 2/10/26.
— Knots in Washington 52. 12/7/25. 20-minute talk.
— UMD Algebraic Geometry Seminar. 12/1/25.
— UT-Dallas Algebra and Combinatorics Seminar. 11/19/25.
— Howard University Mathematics Colloquium. 9/19/25.
— USNA Mathematics Colloquium. 4/17/25.
— Johns Hopkins Topology Seminar. 4/16/25.
— Rice Algebraic Geometry and Number Theory Seminar. 4/1/25.
— UChicago Representation Theory Seminar. 1/7/25.
— Northeastern Geometry, Physics, and Representation Theory Seminar. 10/31/24.
— WUSTL Combinatorics Seminar. 9/30/24.
— AMS Spring Central Sectional Meeting. Special Session on Algebraic Methods in Graph Theory and Applications. 20-minute talk. 4/21/24.
— University at Buffalo Algebra Seminar. 3/4/24.
— Institute for Advanced Study in Mathematics (IASM) Online Geometry and Physics Seminar. Zhejiang University. 1/23/24.
— International Centre for Mathematical Sciences (ICMS) UK–Vietnam Mathematics Joint Meeting. Edinburgh. 12/5/23.
— MIT Lie Groups Seminar. 11/1/23.
— AMS Fall Eastern Sectional Meeting. Special Session on Geometry, Physics and Representation Theory. 50-minute talk. University of Buffalo. 9/10/23.
— Yale Geometry, Symmetry, and Physics Seminar. 4/24/23.
— MIT–Harvard–MSR Combinatorics Seminar. 2/15/23.
— Northeastern Geometry, Physics, and Representation Theory Seminar. 10/6/22.
— ICERM Workshop on Braids in Symplectic and Algebraic Geometry. 10-minute talk. 3/23/22.
— Oxford Algebra Seminar. 11/30/21.
— UMass-Amherst Representation Theory Seminar. 10/25/21.
— Cluster Algebra Seminar. AIM Link Homology research community. 10/7/21.
— Categorical Braid Group Actions and Categorical Representation Theory. UMass-Amherst Focused Research Group (FRG) Conference. 6/17/21.
— MIT Lie Groups Seminar. 3/3/21.

— Chinese University of Hong Kong (CUHK) Representation and Number Theory Seminar. 1/15/21.
— Harvard/MIT Algebraic Geometry Seminar. 9/29/20.
— MIT Lie Groups Seminar. 12/4/19.
— MIT Geometric Representation Theory Seminar. 12/4/19.
— Caltech Algebra and Geometry Seminar. 11/11/19.
— AMS Fall Western Sectional Meeting. Special Session on Geometric Methods in Representation Theory. 20-minute talk. UC-Riverside. 11/9/19.
— UPenn Math-Physics Joint Seminar. 10/22/19.
— Hilbert Schemes, Categorification, and Combinatorics. 30-minute talk. UC-Davis FRG conference. 6/19/19.
— Perimeter Institute Mathematical Physics Seminar. 5/21/19.
— Johns Hopkins Number Theory Seminar. 4/29/19.
— UC-Davis Algebraic Geometry Seminar. 11/14/18.

Non-Research Invited Talks

— Howard University Graduate Student Seminar. 10/31/25.
— Algebraic Geometry Seminar. AIM Link Homology research community. 10/1/21.
— Geometric Categorification Seminar. AIM Link Homology research community. 7/5/21.
— MIT Independent Activities Period (IAP) Mathematics Lecture Series (18.095). 1/20/21.
— Graduate Summer School on the Geometry and Modular Representation Theory of Algebraic Groups. Simons Center for Geometry and Physics (SCGP). 30-minute talk. 8/20/19.
— Interactions between Representation Theory and Algebraic Geometry. 30-minute talk. UChicago. 8/21/17.

Conference and Workshops

— Symmetric Functions and Knot Homology. New Frontiers at the Intersection of Combinatorics and Representation Theory. 3/30–4/3/27.
— Workshop II: Contact Geometry, Cluster Algebras, and Skein Theory. Quantum Topology, Character Varieties and Low-Dimensional Geometry. Institute for Pure & Applied Mathematics (IPAM). 10/26–10/29/26.
— SIAM Texas–Louisiana Sectional Meeting. SMU. 10/30–11/1/26.
— AMS Fall Eastern Sectional Meeting. GWU. 10/3–10/4/26.
— International Congress of Mathematicians (ICM) 2026. Philadelphia. 7/23–7/30/26.
— Summer Workshop on Character Sheaves on Loop Groups. UMinnesota-Twin Cities. 6/8–6/12/26.
— AMS Joint Mathematics Meetings (JMM). Washington, DC. 1/4–1/7/26.
— Webs in Algebra, Geometry, Topology, and Combinatorics. ICERM workshop. 12/8–12/12/25.
— Knots in Washington 52. GWU conference. 12/5–12/7/25.
— Diagrammatic Categorification. ICERM workshop. 10/20–0/24/25.
— Categorification and Symplectic Duality. Northeastern workshop. 6/3–6/7/25.

— Representation Theory Days. MIT conference in honor of George Lusztig. 11/9–11/11/24.
— Summer School and Workshop on Relative Langlands Duality. UMinnesota-Twin Cities. 6/3–6/8/24.
— AMS Spring Central Sectional Meeting. UW-Milwaukee. 4/20–4/21/24.
— Monodromy and Its Applications. Princeton conference in honor of Nick Katz. 12/7–12/9/23.
— UK-Vietnam Mathematics Joint Meeting. ICMS conference. 12/4–12/8/23.
— AMS Fall Eastern Sectional Meeting. University of Buffalo. 9/9–9/10/23.
— Coulomb Branches and Knot Homology. MIT summer school. 6/19–6/23/23.
— Braids in Symplectic and Algebraic Geometry. ICERM workshop. 3/21–3/25/22.
— Link Homology. AIM online research community. 2021–22.
— Categorical Braid Group Actions. UMass FRG conference. 6/14–6/18/21.
— AMS Fall Western Sectional Meeting. UC-Riverside. 11/9–11/10/19.
— Geometry and Representation Theory of Algebraic Groups. SCGP summer school. 8/19–8/23/19.
— Hilbert Schemes, Categorification, and Combinatorics. UC-Davis FRG conference. 6/19–6/23/19.
— Categorified Hecke Algebras, Link Homology, and Hilbert Schemes. AIM workshop. 10/1/18–10/5/18.
— Deligne–Lusztig Theory. Workshop on Algebra and Representation Theory, Held on Oregonian Grounds (WARTHOG) summer school. 8/12–8/18/18.
— Interactions between Representation Theory and Algebraic Geometry. UChicago conference in honor of Sasha Beilinson and Victor Ginzburg. 8/21–8/24/17.
— Soergel Bimodules. MSRI (SLMath) workshop. 6/26–7/7/17.

Teaching

Course Instructor. Howard University 2025–
— MATH 199: Introduction to General Topology (2026)
— MATH 300: Graduate Seminar (2026, *2 lectures*)
— MATH 251: Topology II (2026)
— MATH 250: Topology I (2025, 2026)

Course Instructor. Yale University 2024–25
— MATH 340: Advanced Linear Algebra (2025)
— MATH 430: Introduction to Topology (2025)
— MATH 665: Topics in Quantum Algebra (2024)

Recitation Instructor. MIT 2021–24
— 18.02/18.02A: Calculus (2024, *double recitation*)
— 18.02A: Calculus (2023, *double*)
— 18.01A: Calculus (2023, *double*)
— 18.01A: Calculus (2021)
— 18.02A: Calculus (2021)

Course Instructor. MIT 2022–23
— 18.781: Theory of Numbers (2023)
— 18.708: Topics in Algebra (2022, *4 lectures*)
— 18.704: Seminar in Algebra (2022)

Course Administrator. MIT 2022
— 18.01: Calculus

Graduate Instructor. University of Chicago 2017–20
— MATH 152/44: Calculus II (2020)
— MATH 153/44: Calculus III (2020)
— MATH 132/58: Elementary Functions and Calculus II (2018)
— MATH 131/58: Elementary Functions and Calculus I (2018)
— MATH 195/51: Mathematical Methods for the Social Sciences (2017)

Teaching Assistant. University of Chicago 2015–16
— MATH 242: Algebraic Number Theory (2016)
— MATH 159/53: Introduction to Proof in Linear Algebra and Analysis (2016)
— MATH 175: Elementary Number Theory (2015)

Service

Referee. 2020–

Journals have included *Advances in Mathematics*, *Geometry & Topology*, *International Mathematical Research Notices*, *Journal of the London Mathematical Society*, *Mathematische Annalen*, *Proceedings of the American Mathematical Society*, *Representation Theory*, *Science China Mathematics*, *Transformation Groups*.

Co-organizer. AMS Fall Eastern Sectional Meeting 2026
— Special Session on Springer Fibers, Harmonic Analysis, and Arithmetic

Course Coordinator. Howard University 2026–
— MATH 007: Precalculus

Graduate Admissions Committee. Howard University 2025–

Math Club Faculty Advisor. Howard University 2025–

Co-organizer. GWU–Howard Combinatorics & Algebra Seminar 2025–

Co-organizer. Yale Geometry, Symmetry, and Physics Seminar 2024–25

Research Advisor. PROMYS 2023

Mentor. MIT SPUR 2023
— Maxim Li and Misheel Otgonbayar

Mentor. MIT PRIMES 2021–23
— Michael Yang (2023)
2024 Regeneron STS Scholar

— Luke Robitaille (2021)
2022 Regeneron STS Winner, Tenth Place
2022 Davidson Fellow

Mentor. MIT UROP 2022–23

Research Advisor. MIT PRIMES 2021–23
— Ethan Liu (*Mentor:* Merrick Cai, 2023)
— Joshua Guo (*Mentor:* Kevin Chang, 2021)
2022 Regeneron STS Scholar
— Ilaria Seidel (*Mentor:* Jeffery Yu, 2021)

Mentor. UChicago Directed Reading Program 2015–20

QED Day Judge. Math Circles of Chicago 2019

Mentor. UChicago REU 2016–19
— Karhan Kayan (2019)
— Gwyn Moreland (2016)
— James Stevens (2016)

QED Day Senior Judge. Math Circles of Chicago 2016

Advising Chair. Princeton Math Club 2013

Director. Princeton University Math Competition 2011–12

Honors

— Plenary speaker at Knots in Washington 53.
— **2020 Mathematical Sciences Postdoctoral Research Fellowship (MSPRF)** from the National Science Foundation.
— **2019 William Rainey Harper Dissertation Fellowship** from the University of Chicago.
— 2014 Graduate Research Fellowship (GRFP) from the National Science Foundation.
— Phi Beta Kappa Society at Princeton University.
— Society of Sigma Xi at Princeton University.
— 2011 Shapiro Prize for Academic Excellence from Princeton University.

Profiles

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